Helix VPN — Session Continuation File

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> Helix Constitution §11.4.131 — standing session-resumption artifact. Re-read this file at the start of any new session before touching code. A fresh session can resume with ONLY this file's path plus `.remember/remember.md` — both are kept byte-accurate against live `git rev-parse`/`workable-items validate` output, never against in-context memory or a prior handoff's text alone (see the multi-session lesson below).

ROUND 4: FINAL MVP DOCUMENTATION & PLATFORM READINESS — LANDED

Started: 2026-07-05T11:57:05Z
Landed: 2026-07-06T13:40:00Z
Main repo commit: `d8b9fc1d087bfffa2ba871685bf8ac89687a8740`
Handoff report: `docs/reviews/mvp-final/signoffs/mvp-final-handoff-report.md`

Goal achieved: The consolidated MVP implementation source-of-truth is authored, reviewed, gap-closed, committed, pushed, and verified. The package is ready for development-team kick-off.

Deliverables landed:

1. `docs/research/mvp/final/implementation/` — 13 numbered sections + source-coverage ledger, all with HTML/PDF/DOCX siblings.

2. `docs/design/opendesign/helix/` — OpenDesign design system (manifest, tokens, components, exports); imports cleanly (exit 0).
3. `submodules/helix_proto/` — five `.proto` packages, generated Go stubs, `go.mod`/`go.sum`; `go build ./gen/...` and `buf lint` pass.
4. `submodules/challenges/helix_vpn/` + `submodules/helix_qa/banks/helix_vpn/` — 8 Challenges + 8 HelixQA cases, bidirectionally traceable with the coverage ledger.
5. `docs/research/mvp/final/v04-client/connector.md` — single owning nano-detail doc consolidating FR-701..707.
6. `docs/reviews/mvp-final/review-rounds/round-{1,2}-{docs,design,code,qa}-findings.md` — two full rounds of independent adversarial review.
7. `docs/reviews/mvp-final/signoffs/gap-closure-summary.md` — GAP-1..GAP-6 status, owner, residual, and kick-off blocker flag.

Round-2 adversarial verdicts:

- Docs: **GO-with-conditions** → post-review conditions fixed.
- Design: **GO-with-conditions** → source-manifest consistency fixed; token-contract grade `needs-rebuild` explicitly accepted and documented.
- Code: **GO**.
- QA: **GO**.

Key fixes closed in this round:

- `helix_proto` `go_package` paths aligned with generated directory layout so Go stubs are importable.
- GAP-1 connector local-ACL × central-policy precedence rule backported from the consolidation READMEs into `v03-control-plane/svc-policy.md` and `v04-client/helix-core-rust.md`; `requirements-traceability.md` and `functional-requirements.md` updated to CLOSED/VERIFIED.
- QA coverage ledger made bidirectionally traceable to the actual bank contents; 8 Challenge IDs and 8 HelixQA IDs minted; DDoS entries added for NFR-413/NFR-414.
- OpenDesign source manifest stripped of import-generated path declarations so the source package is reproducible.
- GAP-3 closed at doc level: `v06-deploy/disaster-recovery.md` RTO/RPO targets and runbooks authored; measurement pending Phase-2 CHAOS region-failover drill.
- GAP-4 closed: `v04-client/connector.md` created as the single owning doc for FR-701..707; traceability matrix repointed.
- GAP-5 closed at doc level: all 15 `v08-testing/` nano-detail docs authored and mapped; evidence states honestly PENDING until build.
- GAP-6 closed: DDOS/RBAC/rate-limiting owners pinned; quantitative DDoS targets defined in `v08-testing/ddos.md` §10.

Verification:

- `tests/pre_build_verification.sh` — PASS.
- Mermaid validation — 361/361 blocks rendered ok across `docs/research/mvp/final/`.
- `go build ./gen/...` and `buf lint` in `submodules/helix_proto` — PASS.
- All submodule commits verified by `git rev-parse HEAD` equality against every configured remote.
- Main repo push verified by `git rev-parse HEAD` equality against `github`, `origin`, and `upstream`.

Honest residual measurement gaps (not kick-off blockers): NFR-205 RTO/RPO numbers pending Phase-2 drill; NFR-413/NFR-414 DDoS `ATTACK_PPS` and legit-handshake SLO pending Phase-2 benchmarks; OpenDesign token-contract grade `needs-rebuild` accepted. All are tracked in the coverage ledger, review reports, and handoff report.

ROUND 3.1: FULLY LANDED (2026-07-05T11:29:45Z)

Follow-up clean-up round for deferred items discovered during the Round 3 second decoupling audit. Committed and pushed; verified via direct `git rev-parse` equality against every remote for the main repo AND every touched submodule:

- `docs_chain` — `99ad270` — fixed broken `[Constitution.md](Constitution.md)` self-links in `CONSTITUTION.md` / `AGENTS.md` / `CLAUDE.md` / `QWEN.md`; renamed `Upstreams/` → `upstreams/` and `GitHub.sh/GitLab.sh` → `github.sh/gitlab.sh`; updated `install_upstreams.sh`. Pushed to `origin/github/gitlab/upstream` on `main`.
- `llms_verifier` — `0e7d6949` — fixed broken self-links in `QWEN.md`; lowercase upstream scripts. Pushed to `origin/github/gitlab/upstream` on `main`.
- `panoptic` — `c6b6c49` — fixed broken self-links in `CRUSH.md`; lowercase upstream scripts. Pushed to `origin/github/upstream` on `main`.
- `challenges` — `2711bf0` — added containers-style package metadata table to `README.md`; lowercase upstream scripts. Pushed to `origin/github/ gitlab/upstream/vasicdigitalgithub` on `main`.
- `security` — `318c8c7` — added containers-style package metadata table to `README.md`; lowercase upstream scripts. Pushed to `origin/github/ gitlab/upstream` on `main`.
- `containers` — `df980b3` — lowercase upstream scripts. Pushed to `origin/github/gitlab/upstream/vasicdigitalgitlab` on `main`.

- `doc_processor` — `4e98523` — lowercase upstream scripts. Pushed to `origin/github/gitlab/upstream/vasicdigitalgithub/vasicdigitalgitlab` on `master`.
- `helix_qa` — `04e12e4` — lowercase upstream scripts. Pushed to `origin/github/gitlab/upstream/vasicdigitalgithub` on `main`.
- `llm_orchestrator` — `4aa7219` — lowercase upstream scripts. Pushed to `origin/github/upstream` on `master`.
- `llm_provider` — `084d56f` — lowercase upstream scripts. Pushed to `origin/github/gitlab/upstream/vasicdigitalgithub` on `master`.
- `vision_engine` — `9553a31` — lowercase upstream scripts; rewrote stale `push-all.sh` to iterate all configured remotes and removed four broken single-remote push scripts. Pushed to `origin/github/upstream` on `master`.
- `constitution` — `eae531a` — lowercase upstream scripts. Merged gitflic/main updates before push (no force-push, per §11.4.113). Pushed to `gitflic/ github/gitlab/gitverse/origin/upstream/vasicdigitalgithub/ vasicdigitalgitlab` on `main`.
- Main repo — `ea9677a` — renamed `upstreams/GitHub.sh` → `upstreams/github.sh`; bumped all touched submodule pointers.

Remaining queued (not yet dispatched): `helix_qa`'s nested third-party `tools/opensource/docling` / `tools/opensource/skyvern` working-tree drift (remains untouched; third-party vendored code, exempt per §11.4.28).

Next action: none in flight. Await new operator instructions.

ROUND 3: FULLY LANDED (2026-07-05T07:00:00Z, re-verified this pass)

All Round 3 work is committed and pushed, re-confirmed via direct `git rev-parse` equality against every remote for the main repo AND every touched submodule (never trusted a push-log message or a prior handoff's text alone, per §11.4.88/§11.4.6):

- `helix_core` — `992e1be` (engineering batch), on `origin/main`. Re-verified fresh this pass: `cargo test --workspace` → 120 tests, 0 failed.
- `helix_edge` — `08d6e18` (first real crate), on `origin/main`. Re-verified fresh: `cargo test --all-targets` → 11 tests, 0 failed.
- `helix_go` — `57d4972` (first real Go module + a `.gitignore` fix — a stray `/pkg/` ignore rule was silently excluding `pkg/masqueedge` from version control; caught before push), on `origin/main`. Re-verified fresh: `go build ./...` clean, `go test ./...` → ok.
- `llm_orchestrator` — `ef73c3a` (two-pass CONSTITUTION.md rewrite — a review caught the first pass as incomplete), pushed to `master` on all 3 remotes (this repo's parent-tracked lineage is `master`, not `main`).

- `vision_engine` — `2f22942`, reviewed GO, pushed to `master`.
- `llms_verifier` — `17b4bfb6` (HelixCode + leftover-Lava fix at `9281cae2`, **fast-forwarded** past 3 unrelated upstream commits from a separate workstream — a semantic-code-visibility exit-code fix + a CONST-069 mandate + a reconciliation merge — found and merged during this pass, no force-push, no conflict), on `origin/main`.
- `panoptic` — `31aaceb` (cascaded CONST-048/050/051/052/056 boilerplate fixed on a second pass), on `origin/main`.
- `containers` — `a432efa` (real `os.UserHomeDir()` fix + a synthetic-`$HOME` regression test + 33-package doc-table correction), re-verified.
- `helix_qa` — `c1c2513` (routine nested opensource-tool submodule pointer advancement, confirmed ordinary upstream drift). Its nested third-party tools (`tools/opensource/docling`, `.../skyvern`) show their OWN working-tree drift (a modified test-data file, a modified nested `integrations/n8n` pointer) — deliberately left untouched (third-party vendored code, exempt from equal-engineering per §11.4.28; origin/intent of the drift was not investigated).
- Main repo — `26b4b2a` (bumps `llms_verifier`'s pointer to the fast-forwarded tip above; supersedes the earlier `4d338cb/e96410b/f1de366` pointer-bump sequence). Confirmed identical across `origin/github/upstream` via `git rev-parse`.

Every commit above passed an independent adversarial code-review pass (§11.4.125/§11.4.134) before being accepted — three rounds initially returned NO-GO or hit a mid-review session-limit error (llm_orchestrator's CONSTITUTION.md; the first llms_verifier+panoptic HelixCode pass; the helix_go review) — each was confirmed genuinely completed via git state before being trusted, per the no-guessing mandate, not re-done blindly.

Multi-session lesson (new this pass, keep this note until it's no longer novel): mid-session, this exact working directory was found to have been advanced by a PARALLEL Claude Code session operating the same checkout concurrently — commits landed and `.remember/remember.md` was overwritten with a handoff describing work outside the then-current conversation's own context. The correct response was to treat git state as ground truth and re-verify everything (`git log`, `git rev-parse` against every remote, re-run test suites fresh) rather than trust the handoff text or the in-context conversation memory of "pending work." This generalizes §11.4.37's fetch-before-edit doctrine to a stronger claim: verify against the filesystem even when your OWN memory feels current — a parallel session can invalidate it without warning.

`docs/workable_items.db`: HVPN-P0-018/028/031/039/042/071 closed `Completed (→ Fixed.md)` with real evidence citations. `validate`: PASS, 484 items, 0 issues (re-confirmed this pass).

Remaining queued (not yet dispatched, carried forward to a later round): `helix_qa`'s nested third-party tool drift noted above. The other deferred items (broken `Constitution.md` self-links in `docs_chain`, stale package tables in `challenges/security`, and PascalCase upstream scripts) were dispatched and completed in **Round 3.1** above.

Next action: none in flight. Await new operator instructions (original mandate: notify when fully committed/pushed so the user can send mvp4 instructions — done), or action the queued follow-up round above.

ROUND 3 — Phase-0 Rust/Go engineering + fleet decoupling audit (historical log, landed above)

Round 2 (MVP gap-analysis) + Round 2.1 (decoupling audit) are complete and pushed — see "ROUND 2 + 2.1: FULLY LANDED" below. Round 3 landed real Phase-0 engineering across two submodules plus a second, deeper fleet-wide decoupling audit that surfaced serious findings now being remediated in parallel.

Landed and independently reviewed GO (uncommitted, staged for one batch)

All in `submodules/helix_core` unless noted:

- **HVPN-P0-018/020/021** (orchestrator client/connector binaries) — real WireGuard Noise IK handshake over loopback via `helix-wg`/ boringtun, driven by new `crates/helix-core/src/bin/{helix-client, helix-connector}.rs` + `crates/helix-orch/src/wg_session.rs`. First review returned **NO-GO** (2 findings: a false "nft/iptables not installed" claim — both ARE installed, `nft` runs unprivileged inside an isolated netns; a private key accepted via `--private-key` CLI argv, leaking via `ps`/`/proc/<pid>/cmdline`). Both fixed (env-var-only key resolution via new `cli::read_private_key_from_env`; corrected doc comment/ runtime message to the real constraint — root- owned namespace access to the host's actual LAN NIC, not missing tooling). Re-review: **GO**.
- **HVPN-P0-028/029/030** (quinn+h3 QUIC connection) — `crates/helix-masque/src/quic.rs` taken from stub to real: genuine `quinn::Endpoint` client+server, real hostname/SAN cert verification (no skip-verification shortcut), RFC 9221 datagram round-trip. Review: **GO**.
- **HVPN-P0-031/032/033** (MASQUE CONNECT-UDP + HTTP-Datagram framing) — new `crates/helix-masque/src/{datagram,connect}.rs`. Deep-researched `h3`'s real RFC 9298

support (found genuinely immature — open bugs in its own datagram/quarter-stream-ID handling) and honestly built a labeled simplified stand-in instead of claiming false RFC compliance. Not yet independently reviewed as a standalone item (folded into the same batch as P0-018/028, which were).

- **HVPN-P0-071/072/073** (map.json schema + reconciler) — new `crates/helix-core/src/map.rs`. Pure diff engine, idempotent, panic- free on adversarial malformed/duplicate-peer inputs. Review: **GO** (two non-blocking follow-ups noted for Phase-1: canonicalize `allowed_ips` ordering before live wiring; add an explicit CONC/RACE test per the WBS's own declared test-types).
- **HVPN-P0-042/043/044** (Go edge, `quic-go`+`masque-go`) — bootstrapped `submodules/helix_go`'s first real Go module (`pkg/masqueedge` + `cmd/go-edge`). `masque-go` proved genuinely turnkey (real CONNECT-UDP server+client wired from its own test-suite pattern, no hand-rolled framing needed) — concrete evidence for the Go-vs-Rust edge-language decision, in Go's favor for this specific protocol layer. Not yet independently reviewed.
- **HVPN-P0-039/040/041** (Rust edge `helix-edge`) — bootstrapped `submodules/helix_edge`'s first real Cargo binary, path-depending on the sibling `helix_core` crates. A **complete real WireGuard handshake traverses MASQUE-client → edge-relay → `helix_orch::wg_session` responder**. A genuine bug was found and fixed during TDD: `send_datagram()` only enqueues (doesn't flush) — an early test closed the connection immediately after, racing quinn's async flush and dropping the handshake's 3rd message; fixed by removing the premature close, stable across 5 reruns. A decoy HTML responder coexists with the real MASQUE flow on the same port number (TCP vs UDP/QUIC — independent kernel namespaces). Review in progress.
- Fixed a real, pre-existing (predates this session, from commit `405db88`) decoupling violation the earlier audit missed: `helix-masque`'s `MasqueConfig`/`QuicConfig` default values hardcoded `proxy.helixvpn.io` — directly contradicting the submodule's own README/CLAUDE.md claim of "no HelixVPN-specific hostnames." Fixed to the generic `proxy.example` (RFC 2606) across all 6 occurrences; 27/27 `helix-masque` tests still pass.
- Fixed a bare-pinned `hex = "0.4"` dependency (used identically by two crates, `helix-wg` and `helix-core`) into `[workspace.dependencies]` per this project's own written convention ("never a separately- pinned version") — flagged by the map.rs reviewer as a minor drive-by finding from the P0-018 batch.

Honest environment constraint (re-confirmed multiple times this round via direct probes, never assumed): no passwordless sudo (`sudo -n true` fails) → real kernel WireGuard, real network namespaces reaching the host's actual LAN interface, and binding privileged port 443 are NOT autonomously achievable here. Every item above is honestly scoped to what's provable via loopback + unprivileged high ports, exactly as this project's own anti-bluff discipline requires — no faked privilege, no silently-skipped acceptance criteria.

Second fleet-wide decoupling audit — severe findings, remediation in progress

A deeper audit (beyond Round 2.1's pass) found:

1. **Wrong-project contamination (worse than the earlier "Lava §6.AD" dangling-sentence bug)**: `llm_orchestrator`, `vision_engine`, `llms_verifier`, `panoptic` have their ENTIRE `CLAUDE.md`/`AGENTS.md` /`CONSTITUTION.md` bodies describing a different, unrelated project called "HelixCode" (22-62 grep hits each) — nonexistent directories, wrong package structures, wrong Makefile targets, wrong module names. Each submodule's own `README.md` is correct and was used as ground truth for the fix. Remediation dispatched (2 parallel subagents, one per submodule pair).
2. **`llms_verifier` still carries leftover "Lava"-project content** beyond what the earlier session-wide fix removed — a full section ("§6.X — Container-Submodule Emulator Wiring Mandate") explicitly referencing "the parent Lava repo." Being removed/genericized as part of the same remediation pass.
3. **`containers/pkg/remote/compose_detector.go:76` hardcodes this operator's home directory in PRODUCTION CODE** (not just docs) as a `podman-compose` lookup candidate — silently never matches on any other machine. Fix dispatched: real `os.UserHomeDir()` resolution + a test that would have caught the original bug.
4. Broken `[Constitution.md](Constitution.md)` self-links (should be `CONSTITUTION.md`, uppercase) in `docs_chain`/`llms_verifier`/`panoptic`; stale package tables in `challenges`/`containers`/ `security` (missing 3-16 real packages each); PascalCase `GitHub.sh`/`GitLab.sh` upstream scripts in 11 "borrowed" submodules (violates §11.4.29 lowercase-snake_case) — queued for a follow-up round, not yet dispatched.

Next actions — all of Round 3's own steps are DONE; see "Remaining

queued" above for the one carried-forward follow-up round.

Milestones S4-S8's remaining subtasks (bench.sh A/B harness, decoy/ DPI-survival gates, FFI/mobile work) remain `Queued` — most are gated behind the edge-implementation work above, which has now landed and reviewed cleanly, so these are unblocked for a future round.

ROUND 2 + 2.1: FULLY LANDED (2026-07-05T00:14:23Z, verified)

Both main-repo commits are confirmed pushed to all 3 remotes — local HEAD, `github/main`, `origin/main`, `upstream/main` all equal `20dc9a4` (verified via `git rev-parse`, not assumed). Round 2's first push attempt (`e46a710`) stalled/died across a session/process restart mid-transfer (~84MB of new binary content) and had to be retried from scratch — the retry completed in seconds, confirming the first attempt was genuinely stuck, not just slow. **Lesson for future large pushes in this environment: verify completion by comparing `git rev-parse HEAD` against every remote's tip, never trust a "push started in background" message alone — nohup+disown does not reliably survive a session/process restart in this environment.**

All 19 owned submodules verified in sync with their own remotes (15 confirmed byte-identical local==origin; the 4 known-diverged ones —doc_processor/llm_orchestrator/llm_provider/vision_engine — correctly left untouched at the main-repo-pointer level, with llm_provider's actual fix independently confirmed live on its real remote).

Round 2 + 2.1 are now genuinely, verifiably complete — commits pushed, submodule pointers current, nothing pending.

ROUND 2.1 — post-completion cleanup (2026-07-04T18:10-19:30, DONE, LANDED)

After round 2 was declared complete, two more real, operator-directed fixes landed across the submodule fleet (main repo's own commit for this is NOT yet made — main repo's huge round-2 push is still uploading; the pointer bumps for these submodules will land in the NEXT main-repo commit):

1. **"Lava §6.AD" dangling cross-project reference** — removed from 19 submodules' `CLAUDE.md`/`AGENTS.md`/`CONSTITUTION.md` (a stray sentence claiming this project's root CLAUDE.md has a "§6.AD" about incorporating an unrelated project called "Lava" — it does not). **Important sub-finding**: 4 submodules (`doc_processor`, `llm_orchestrator`, `llm_provider`, `vision_engine`) have a LOCAL checkout on an independently-diverged `master`-lineage branch vs. their `origin/main` — confirmed via `merge-base --is-ancestor` (NOT corruption; operator confirmed this is expected/known). Fixed directly on `origin/main` (the branch that matters) via a clean detached worktree for the one repo that needed it there (`llm_provider` — the other 3's `main`

was already clean of this specific issue); did NOT touch/merge/push the diverged `master` lineage for any of the 4. `llm_provider`'s `vasic-digital` mirror (github+gitlab) additionally rejected the push as non-fast-forward (yet another independent divergence) — left untouched, same "expected divergence" class per operator confirmation.
2. **Decoupling violation — hardcoded "Helix VPN" project name** — found (operator mandate: "Keep all Submodules fully decoupled! No Submodule can be parent project aware!!!") in the 8 freshly-created submodules' governance files AND in `helix_core`'s REAL source metadata (`Cargo.toml description` fields, `lib.rs` doc comments, `README.md` — pre-existing from an earlier session, not this round's governance work). Fixed all of it generically (e.g. "a consuming VPN/networking product"); verified `cargo check --workspace` still compiles clean (0 new warnings) after the text-only changes. Comprehensive fleet-wide `git grep` sweep across all 19 submodules' TRACKED files confirms zero remaining "helix vpn"/"helix_vpn" references anywhere.

All 9 affected submodules (`docs_chain` + the 8 fresh ones, plus `llm_provider`'s main separately) committed + pushed to their primary remotes. **Main repo's submodule-pointer bump for all of this is still pending** — do not forget it in the next main-repo commit.

ROUND 2 STATUS: COMPLETE (2026-07-04T18:10:00Z)

MVP gap-analysis/hardening round 2 (see full detail retained below) is **DONE**: all 3 MVP corpora hardened + unified, independent review gate passed (after a real fix cycle — 8 critical findings fixed, not rubber-stamped), governance propagated to all 19 owned submodules (9 newly fixed, all committed+pushed to their own remotes), workable-items DB reconciled with real `cargo test` evidence, all exports regenerated (0 failures after fixing 4 real mermaid defects + 1 Puppeteer/Chrome launch bug), main repo committed (`feat: MVP gap-analysis + enterprise hardening round...`), push to all main-repo upstreams running detached per §11.4.88 (check `qa-results/push_failures/` for any failure log — absence = success). **Task #6 (anti-bluff constitution propagation) is also now COMPLETE**: root project was already compliant; all 19 owned submodules now carry proper CLAUDE.md/AGENTS.md/QWEN.md/CONSTITUTION.md with the inheritance pattern; the project's real Rust tests were confirmed genuine (not mocked) during the DB-reconciliation work. **Nothing from this round's scope remains open** except the explicitly-flagged, intentionally out-of-scope items in §5 below (operator decisions) and the low-priority "Lava §6.AD" fleet-wide cleanup item (tracked, not urgent).

If resuming fresh: verify the push actually succeeded (`git log --oneline HEAD..@{u}` for each remote should be empty, or check `qa-results/push_failures/` for a failure log) before assuming this round is 100% externally visible — the commit itself is durable either way.

CURRENT ROUND (2026-07-04, round 2) — MVP gap-analysis, hardening, unification

Operator mandate (verbatim intent): analyze all three MVP corpora (`docs/research/mvp/`, `docs/research/mvp2/`, `docs/research/mvp3/` + `docs/research/mvp_final/`) for gaps / inconsistencies / shortcomings / unfinished parts; close every gap with rock-solid, enterprise-grade, scalable content; extend all docs/guides/plans/diagrams/OpenDesign cross-references to make all MVP phases impeccable; commit + push everything (main + all submodules, all upstreams); keep CONTINUATION.md + exports in sync throughout; use subagent-driven parallel execution; notify when fully committed/pushed so the operator can send mvp4 instructions. **Follow-up mandate queued right behind this one** (do NOT drop it — see "Queued follow-up" below): propagate the anti-bluff testing/Challenges mandate into this project's `Constitution.md`/`CLAUDE.md`/`AGENTS.md`/`QWEN.md` and every owned submodule's equivalents, respecting the HelixConstitution submodule inheritance rules (§11.4.35/§11.4.26).

Status: 4 parallel subagents dispatched, IN PROGRESS (not yet returned as of this write):

1. **Agent A** — deep gap analysis + direct hardening of `docs/research/mvp/` (Phase 0/1 control-plane corpus — note the actual numbered-volume spec lives under `docs/research/mvp/final/*.md`, plus `docs/research/mvp/04_VPN_CLD/*.md`; agent was told to `find docs/research/mvp -name '*.md'` first so it discovers the real paths itself).
2. **Agent B** — deep gap analysis + direct hardening of `docs/research/mvp2/` (Phase 2, 8-platform client-app corpus).
3. **Agent C** — defines Phase 3 (`docs/research/mvp3/MVP3_ENTERPRISE_SCALE.md`, new) and the GA/Final phase (`docs/research/mvp_final/MVP_FINAL_GA_READINESS.md`, new) from scratch (both were previously empty `TBD.md` placeholders), plus a new unifying `docs/research/UNIFIED_PHASE_ROADMAP.md` reconciling all phases into one index.

4. **Agent D** — cross-cutting, read-only-on-others audit: `docs/research/CROSS_CUTTING_GAP_ANALYSIS.md` (new) covering OpenDesign coverage across mvp2's 8 platforms, testing-philosophy consistency between mvp/ and mvp2/, and a diagram-completeness sweep across all corpora — produces recommendations only, does not edit mvp/mvp2/mvp3/mvp_final/design directly (disjoint file-scope from Agents A/B/C per §11.4.58/§11.4.20).

Next actions once all 4 agents return (in priority order):

1. Review each agent's final report; spot-check the diffs for quality/consistency (§11.4.92 multi-pass — at minimum Pass 1 main-task + Pass 2 blast-radius).
2. Action Agent D's recommendations that are cheap/obvious (e.g. missing OpenDesign cross-refs in mvp2 UI/UX spec) — either inline now or track as a fast-follow.
3. Re-sync exports: run whatever markdown→HTML/PDF/DOCX export pipeline this project uses for `docs/research/**` (check `scripts/testing/sync_all_markdown_exports.sh` — confirm it still exists and covers the new files before invoking) — per §11.4.65/§11.4.12, exports must stay in sync with sources.
4. If `docs/workable_items.db` / Issues.md tracking is meant to cover this round's work, reconcile per §11.4.93/§11.4.148 (check whether this project's tracker actually requires per-doc workable items for a docs-only round, or whether that's overkill here — do not force-fit a heavyweight tracker entry for a pure documentation-authoring pass unless the project's own convention already does so).
5. Commit + push: main repo via `scripts/commit_all.sh` (NOT raw `git commit`), and any owned submodule that was touched (none expected this round — all 4 agents were scoped to `docs/research/**`, which lives in the main repo, not a submodule) — push to all configured upstreams.
6. Report completion to the operator with a summary of every file created/modified.
7. **Then** address the queued follow-up mandate below (anti-bluff Constitution/CLAUDE.md/AGENTS.md/QWEN.md propagation) — do not let it drop; it was explicitly deferred, not abandoned.

Workable-items DB reconciliation (DONE this round — real evidence, not a guess)

The DB (`docs/workable_items.db`, 484 items) was found 100% `Queued` — stale against reality: `submodules/helix_core` is actually AHEAD of what this file previously described. Verified with real commands (not trusted from old notes): `cargo test --workspace` inside `submodules/helix_core` (HEAD `405db88` — "WireGuard boringtun + orchestrator + MASQUE stub + G1 tests"), crate source inspection for stub/TODO markers, and workspace `Cargo.toml` inspection. **Actual current state: 6 crates** (`helix-core`, `helix-

`masque`, `helix-orch`, `helix-transport`, `helix-tun`, `helix-wg` — not 4), **72 unit/integration tests passing** (helix_masque 12, helix_orch 13, helix_transport 12, `tests/g1_integration.rs` 3, helix_tun 5, helix_wg 27; plus doctests) — not the "39 tests" this file previously claimed. Using the canonical `cmd/workable-items` Go CLI (`close --id ... --status ... --evidence ...`, never raw SQL), closed with cited evidence:

- **Completed (→ Fixed.md):** HVPN-P0-001 (workspace bootstrap), HVPN-P0-004 (Transport trait, 12 tests), HVPN-P0-008 (UDP transport, proven via G1 echo test), HVPN-P0-011 (real `boringtun=0.7.1` wrapper — genuine dependency + `Tunn`-based device/handshake/noise/timers, zero stub markers, 27 tests), HVPN-P0-015 (TUN device, 5+1 tests), HVPN-P0-022 (test rig, referenced directly by the G1 integration test's doc comment), HVPN-P0-025 (G1 gate — `g1_udp_loopback_echo` really round-trips 10 UDP datagrams with RTT assertions, 3/3 passing).
- **In progress (honest, NOT overclaimed):** HVPN-P0-018 (orchestrator three-loop core is real and tested — 393+226 lines, 13 tests — but zero `[[bin]]` targets exist yet, so the task's own "client/connector binaries" deliverable is unmet), HVPN-P0-028 + HVPN-P0-031 (the `helix-masque` crate's OWN doc comment says "This is a **research / stub pass**" — honored that self-assessment rather than closing).
- DB re-`validated` clean (484/484, 0 issues) after every update; WAL-checkpointed per §11.4.95. This DB change + the corrected crate/test counts above still need to be committed alongside the rest of this round's work (see "Next actions" below — nothing has been committed yet as of this write).

Independent-review gate + fix cycle (2026-07-04 ~17:00-18:00, DONE)

Per §11.4.125/§11.4.134 (code-review-before-build, iterate-until-clean-GO), dispatched 3 independent (structurally-separated, not self-review) adversarial review agents against everything this round produced, before considering it committable:

1. Review of the 5 new top-level docs (`mvp3/MVP3_ENTERPRISE_SCALE.md`, `mvp_final/MVP_FINAL_GA_READINESS.md`, `UNIFIED_PHASE_ROADMAP.md`, `GOVERNANCE_INHERITANCE_AUDIT.md`, `CROSS_CUTTING_GAP_ANALYSIS.md`) — found 3 wrong section-number citations in `MVP3_ENTERPRISE_SCALE.md` (§7.2 cited instead of §4 twice, §2.1 cited instead of §1.2 once) + confirmed both audit docs' headline findings had already been remediated by parallel work (accurate when written, stale by the time of review).
2. Review of the mvp/+mvp2/ hardening diffs — found 2 real defects in mvp/ (a broken relative link the pass itself introduced in `v01-product/functional-requirements.md`; a stale

"still needs fixing" note in `08-phase2-parity-wbs.md` describing a fix that had ALREADY landed in the same diff) and 3 in mvp2/ (a pre-existing broken table row in `MVP2_OVERVIEW.md` §9.2 left unfixed despite heavy editing; an unescaped pipe in `MVP2_SECURITY_PERFORMANCE.md` breaking a table; the self-flagged Phase-1-"COMPLETED" contradiction in `MVP2_OVERVIEW.md` §3.1 that `MVP2_WEB_CLIENT.md` had named but never actually annotated in the source file itself) + 1 minor (residual semantic-color drift in `MVP2_MOBILE_APPS.md` beyond just the primary/accent brand color).
3. Review of the 9-submodule governance propagation — **clean pass, zero critical findings** (already committed/pushed, see below).

All Critical + the one worthwhile Minor finding were fixed directly (myself, not delegated — small, well-specified, mechanical changes) and each fix was empirically verified, not just asserted: the broken link now resolves (`test -f` confirmed); the stale note now says "already fixed" with the real evidence; the `MVP2_OVERVIEW.md` table row now has the correct cell count (verified via `awk -F'|'`); the pipe is escaped; the Phase-1 contradiction now carries an explicit flag note (matching the precedent `UNIFIED_PHASE_ROADMAP.md` R-2 already set — flag, don't silently rewrite); the 3 mvp3 citations point at the confirmed-correct sections; the two now-stale audit docs (`GOVERNANCE_INHERITANCE_AUDIT.md`, `CROSS_CUTTING_GAP_ANALYSIS.md`) each got a "STATUS UPDATE" callout so a future reader doesn't re-do already-closed work; the semantic colors in `MVP2_MOBILE_APPS.md` now cite `color.json`'s real hex values.

Bonus, found independently of the 3 review agents, during my own export-sync QA pass: 3 MORE genuine mermaid syntax defects the review agents' text-only reading couldn't have caught (they don't render diagrams) — all root-caused via direct `mmdc` reproduction + bisection, never guessed: (a) `07-phase1-mvp-wbs.md` — a literal `;` inside a sequence-diagram message breaks mermaid's parser regardless of arrow syntax; (b) `MVP2_WEB_CLIENT.md` — an unquoted flowchart decision-node label containing `(...)` needs quoting; (c) `MVP2_SHARED_CORE.md` — mermaid's `stateDiagram-v2` transition- label parser cannot handle a literal `::` (confirmed via a 6-way bisection test matrix, not assumed); (d) `MVP3_ENTERPRISE_SCALE.md` — the `A -. label .-> B` dotted-arrow-with-inline-label form isn't valid mermaid; the pipe-delimited `A -.->|"label"| B` form is. All 4 verified via actual successful PNG render before moving on, all 11 affected files' exports re-synced with `--force` (0 failures).

This is now genuinely done — not just self-reported by the authoring agents, but independently adversarially reviewed AND the findings fixed AND re-verified. Ready to commit.

Session-limit crash + recovery (2026-07-04 ~16:16 MSK)

Agents A (mvp/) and B (mvp2/, plus B's own 3 sub-dispatched helpers for `MVP2_WEB_CLIENT.md`/`MVP2_MOBILE_APPS.md`/ `MVP2_IMPLEMENTATION_ROADMAP.md`) all failed simultaneously with "session limit resets 4:10pm (Europe/Moscow)" — a hard external wall, not a design/logic bug. NO WORK WAS LOST: their edits already landed on disk before the crash (108 files / +2430 lines in `mvp/`, all 10 `MVP2_*.md` + siblings / +4791 lines in `mvp2/` — verified via `git diff --stat`, not assumed). Confirmed the reset had already passed by the time of investigation and dispatched 2 continuation agents (NOT a repeat of the original giant prompts — targeted at (a) the SPECIFIC dangling thread each crashed agent was mid-sentence on when cut off, cited verbatim from their last output, and (b) reconciling 3 unactioned findings from `CROSS_CUTTING_GAP_ANALYSIS.md`: brand-color inconsistency across mvp2/docs/design, missing anti-bluff testing philosophy in mvp2/, thin diagram coverage in mvp2/). Both launched without error — confirms the limit has cleared. **If a fresh session picks this up and finds these still running or newly crashed again, re-dispatch using the SAME targeted-continuation pattern (not the original full prompts) — check `git diff --stat` first to see exactly what's already done.**

Governance-file propagation (DONE + already committed/pushed — the ONE piece of this round that IS already on remotes)

The 8 zero-governance submodules (`helix_core`, `helix_design`, `helix_edge`, `helix_go`, `helix_proto`, `helix_shims`, `helix_transport`, `helix_ui`) each got `CLAUDE.md`/`AGENTS.md`/ `QWEN.md`/`CONSTITUTION.md`, replicating the exact pattern already used by compliant siblings (`security`, `challenges`) — verified against those two + spot-checked against `docs_chain`/`vision_engine`/ `llm_orchestrator` before writing anything. `panoptic` got its missing `QWEN.md` + an inheritance pointer added to its existing `CONSTITUTION.md`. **All 9 submodules were committed + pushed to their own remotes already** (each is an independent git repo with its own `origin`; `panoptic` additionally has `github`/`upstream` remotes, all three identical URLs, pushed to all three). Commit hashes: helix_core `f245b98`, helix_design `7fbe145`, helix_edge `3c7771a`, helix_go `1eccd10`, helix_proto `24b41a0`, helix_shims `e7c6b43`, helix_transport `cdce305`, helix_ui `bd2a495`, panoptic `8212dbe`.

A flagged-but-deliberately-NOT-fixed finding from that pass: a stray "Lava §6.AD" sentence is duplicated verbatim across ALL previously- compliant submodules' governance files (a pre-existing copy-paste artifact from some unrelated project, predating this round) — it was replicated into the 9 newly-touched submodules too, for fleet consistency, rather than silently patched out mid-round. This is a real, separate, low-priority cleanup item — track it, don't drop it, but it is NOT part of the current round's scope.

IMPORTANT — main repo NOT yet committed for this. The main repo's own commit (which needs to bump all 9 submodule pointers, plus land the workable-items DB reconciliation + this file's edits + GOVERNANCE_INHERITANCE_AUDIT.md/ CROSS_CUTTING_GAP_ANALYSIS.md/ UNIFIED_PHASE_ROADMAP.md/mvp3+mvp_final content) is being held back on PURPOSE: `scripts/commit_all.sh` does `git add -A`, and Agents A (mvp/ hardening) and B (mvp2/ hardening) are STILL actively writing to tracked files as of this write — running the full-repo commit wrapper right now would risk sweeping up their in-progress, possibly-incomplete edits (§11.4.84 working-tree-quiescence concern, generalized from mutation-gates to any concurrent subagent write). **Wait for A and B to fully complete before running `scripts/commit_all.sh` for the main repo.**

Queued follow-up (do not drop): anti-bluff testing/ Challenges mandate propagation

The operator's anti-bluff covenant (tests + Challenges MUST prove real end-user-usable functionality, not just green CI — verbatim historical anchor already lives in `constitution/CLAUDE.md` §11.4 family, esp. §11.4.1/.2/.5/.6/.27/.50/.52/.69/.98/.107/.123/.134/.142) needs to be confirmed present (or added) in THIS project's own root `Constitution.md/CLAUDE.md/AGENTS.md/QWEN.md` if this project maintains project-level copies distinct from the inherited `constitution/` submodule, AND cascaded to every owned submodule under `submodules/*` per §11.4.28/§11.4.35 inheritance rules — universal content belongs in the `constitution/` submodule (already there); project-specific restatement/pointer belongs in this repo's own governance files per the inheritance pattern in `constitution/CLAUDE.md` "How inheritance works". Action: audit whether helix_vpn's own `CLAUDE.md/AGENTS.md` (this repo's root files, not the submodule) already carry the required inheritance pointer + anti-bluff restatement; if any owned submodule (`submodules/helix_core`, `helix_design`, `helix_edge`, `helix_go`, `helix_proto`, `helix_qa`, `helix_shims`, `helix_transport`, `helix_ui`, `llm_orchestrator`, `llm_provider`, `llms_verifier`, `panoptic`, `security`) is missing its own constitution submodule inheritance, fix per §11.4.26 (fetch+pull constitution submodule first, in EACH affected submodule, before editing).

Summary

Branch: `main` (single working branch). **Overall status:** Full MVP specification set COMPLETE. **Design System COMPLETE** (26 files, ~6,700 LOC). **Phase 0 Implementation ADVANCED** — 6 Rust crates, 72 unit/integration tests passing (corrected 2026-07-04 — see "Workable-items DB reconciliation"), submodule pushed. **MVP gap-analysis/hardening round 2 IN PROGRESS** (see "CURRENT ROUND" above).

Active work (2026-07-04):

1. MVP spec set — 11 vols, 126 md/html/pdf, all synced
2. Constitution — fully integrated, pre-commit hooks active
3. Design System — OpenDesign, 30+ components, 18+ screens, 4 exports
4. docs_chain — 3 contexts, all doctor PASS
5. P0-001: Workspace skeleton — 4 crates, compiled
6. P0-022: Test rig — 7 scripts, netns+nftables+netem
7. P0-080/-077: Make + Bench + Spike — 11 targets
8. P0-004: Transport trait refinement — close(), local_addr(), peer_addr(), mock transport
9. P0-008: Plain-UDP transport — UdpTransport, UdpConnection, 12 tests
10. P0-015: Linux TUN device — helix-tun crate, 5+1 tests
11. P0-011 prep: WireGuard stub — helix-wg crate, 5 modules, 21 tests
12. Workspace: 39 tests total — 4 crates, 0 failures, 0 errors
13. Design review fixes — F8, F10, F11, PDFs re-exported
14. All committed + helix_core submodule advanced

Next work queue (corrected 2026-07-04, round 2 — see "Workable-items DB reconciliation" below; P0-011/-018/-025 were already substantively done and are no longer next-up):

1. HVPN-P0-018 remainder: client/connector `[[bin]]` targets wired to the already-implemented orchestrator core (the core loops + 13 tests exist; the binaries do not yet)
2. HVPN-P0-028/HVPN-P0-031: take `helix-masque` from its self-declared "research / stub pass" to production-grade QUIC/MASQUE (needed for G2)
3. Platform adapters — VpnService, NEPacketTunnelProvider, WFP

Locations: spec: `docs/research/mvp/final/` | design: `docs/design/` | Rust: `submodules/helix_core/` (4 crates, 39 tests) | rig: `scripts/rig/`

Completed Work (highlights)

1. Constitution submodule + mandatory submodules

- `constitution/` → `HelixDevelopment/HelixConstitution.git` (branch `main`)
- 11 own-org repos under `submodules/<name>` (flat, lowercase snake_case)
- `install_upstreams` run in each; `.helix-manifest.yaml` audit record
- Pre-commit hook, CI DISABLED (§11.4.156), local enforcement active

2. Full MVP specification set (V0–V10)

- 11 volumes, ~140 nano-detail documents, ~11.7K lines in the spine + pass-1 set
- All 16 research docs cited; decisions D1–D8 surfaced
- 46 Mermaid diagrams, SQL DDL, Podman/Docker/K8s manifests
- Every volume adversarial-reviewed (§11.4.142) + reconciled to GO (§11.4.134)
- 126 `.md` / 126 `.html` / 126 `.pdf` — all synced (§11.4.65)

3. Workable-items SQLite DB (§11.4.93)

- `docs/workable_items.db` — 484 items (P0: 36, P1: 210, P2: 132, P3: 96)
- Schema: items, item_history, test_diary, gates, operator_block_details, obsolete_details, meta
- Loader: `scripts/workable_items_loader.py` (md-to-db, bidirectional)
- All items start as `Queued` / `Task` status

4. Research corpus

- `docs/research/mvp/` — 16 source docs (11 LLM analyses + 5 refined)
- `v09-research/` — 10 per-angle research dossiers (all cited, all verified except wireguard partially)

What Remains

Done (all subagents completed)

- **D-PKI-CA-TIER** — operator confirmed: two-tier issuing CA as MVP default
- **D-OD-1** — operator confirmed: OpenDesign authoring-layer interpretation
- **vasic-digital component repos** — 8 repos created on GitHub+GitLab + added as submodules
- **Go workable-items binary** — HVPN-P1-150 complete, 6 commands verified
- **DOCX exports** — pipeline updated, all docs have DOCX siblings
- **Design System COMPLETE** — 26 files, ~6,700 LOC
  - OpenDesign 9-section DESIGN.md with light+dark themes + 5 custom palettes
  - tokens.css (200+ CSS custom properties) + Figma Variables-compatible JSON
  - Component library (30+ components, 4 platform variants)
  - Screen wireframes (18+ screens across 8 platforms)
  - Interaction patterns + animation specs
  - Exports: 4 PDF, 4 HTML, 2 PNG screenshots
- **Phase 0 Implementation ADVANCED** — 4 Rust crates, 39 tests, all pushed
  - helix-transport: Transport trait + UDP transport (12 tests)
  - helix-tun: async Linux TUN device (5+1 tests)
  - helix-wg: WireGuard stub + timers (21 tests)
  - helix-core: workspace re-export (0 tests)
- **Phase 0 Implementation — Test Rig** (HVPN-P0-022)
  - 7 scripts (common, setup, teardown, test_reach, test_firewall, test_netem, README)
  - 3-namespace topology (client/bridge/server) with nftables + netem
  - G1 precondition gate scriptable
- **Phase 0 Implementation — Infra** (HVPN-P0-080/-077)
  - Makefile with 11 targets (spike, check, test, bench, rig, clean, etc.)
  - scripts/spike.sh (S0→S4 one-shot verification command)
  - scripts/bench/run.sh + compare.sh (iperf3/ping, CSV output)
- **Design quality review** — 15 findings, 5 fixed (F1,F2,F4,F7,F13)
- **Docs chain** — 'design' context registered (12 nodes, 8 edges, doctor PASS)
- **Submodule pushes** — helix_core (first Rust code), containers (exec fixes)

Known issues

- `install_upstreams` recipe format mismatch: recipe files use `GIT_SSH_URL` but the script expects `UPSTREAMABLE_REPOSITORY`. Remotes configured manually. Should be fixed upstream in the Upstreamable toolkit.
- `helix_qa` nested submodules (docling) still dirty — pre-existing, not from our work
- `docs_chain` submodule has dirty tracked file — pre-existing, needs upstream fix
- Design system: OpenDesign CLI (`od`) is GNU octal dump, not the OpenDesign tool — no local OpenDesign agent for automated Figma generation

Deferred

- **Phase 0 Remaining (HIGH)** — P0-011 (boringtun wire), P0-018 (orchestrator three-loop), P0-025 (G1 test with rig), P0-028 (QUIC/MASQUE)
- **Figma design file generation** — requires OpenDesign CLI install or Figma MCP authentication
- **UI implementation** — requires core transport layer stable first
- **Platform adapters** — Android VpnService, iOS NEPacketTunnelProvider, Windows WFP, Linux nftables — each needs helix_core FFI stable

---

Evidence Locations

Artifact	Path
MVP spec set	`docs/research/mvp/final/` (126 md/html/pdf)
Master index	`docs/research/mvp/final/MASTER_INDEX.md`
Spec spine	`docs/research/mvp/final/SPECIFICATION.md`
Research corpus	`docs/research/mvp/` (16 source docs)
Research dossiers	`docs/research/mvp/final/v09-research/`
Workable-items DB	`docs/workable_items.db` (§11.4.93/.95)
DB loader	`scripts/workable_items_loader.py`
DB loader docs	`docs/scripts/workable_items_loader.md`
docs_chain wrapper	`scripts/docs_chain_md_to_db.sh`
docs_chain contexts	`.docs_chain/contexts/*.yaml`
.gitignore-meta	`.gitignore-meta/*.yaml` (§11.4.77 regen mechanisms)

Artifact	Path
Pre-build gate	`tests/pre_build_verification.sh` (8 invariants)
Export script	`scripts/testing/sync_all_markdown_exports.sh`
Mermaid helper	`scripts/testing/render_mermaid_blocks.py`
Mermaid cache	`.mermaid-cache/` (content-addressed PNGs)
Constitution	`constitution/` (submodule)
Submodule audit	`.helix-manifest.yaml`
Pre-commit hook	`.githooks/pre-commit`
CI (DISABLED)	`.github/workflows/constitution.yml.disabled-local-only`
DESIGN SYSTEM	**`docs/design/`** (26 files, ~6,700 LOC)
OpenDesign DESIGN.md	`docs/design/opendesign/helix/DESIGN.md`
OpenDesign tokens.css	`docs/design/opendesign/helix/tokens.css`
OpenDesign manifest	`docs/design/opendesign/helix/manifest.json`
Component reference	`docs/design/opendesign/helix/components.html`
Component library doc	`docs/design/components/README.md`
Screen wireframes	`docs/design/screens/README.md`
Interaction/ animation	`docs/design/interaction/README.md`
Design master index	`docs/design/README.md`
Color tokens JSON	`docs/design/tokens/color.json`
Typography tokens	`docs/design/tokens/typography.json`
Figma tokens JSON	`docs/design/exports/HelixVPN-Figma-Tokens.json`
Design export PDFs	`docs/design/exports/HelixVPN-*.pdf` (4 files)
Design screenshots	`docs/design/exports/HelixVPN-Components-*.png` (2 files)
Platform-specific	`docs/design/components/{desktop,mobile,aurora,web}/*.md`

Resumption prompt (§11.4.127)

SHORT variant

> Continue work on `main` in `/run/media/milosvasic/DATA4TB/Projects/helix_vpn`; read `docs/CONTINUATION.md` first (esp. "ROUND 3: FULLY LANDED" at the top — Rounds 1-3 are all complete and pushed). Pick up the queued follow-up round: broken `Constitution.md` self-links in `docs_chain`, stale package tables in `challenges/security`, and PascalCase `GitHub.sh`/`GitLab.sh` upstream scripts in 11 submodules (§11.4.29) — or await new operator instructions (mvp4).

FULL variant

```
You are resuming work on the Helix VPN project.

Repository:  /run/media/milosvasic/DATA4TB/Projects/helix_vpn
Branch:      main
Handoff doc: docs/CONTINUATION.md  ← read this FIRST, especially
             "ROUND 3: FULLY LANDED" at the top.

State at handoff (2026-07-05, Rounds 1-3 all complete and pushed)
--------------------------------------------------------------------
- Round 1 (complete): full MVP spec set, OpenDesign system,
Phase-0
  Rust scaffolding, workable-items DB.
- Round 2 (complete): full gap-analysis + enterprise hardening
across
  mvp/mvp2/mvp3/final docs, unified phase roadmap. Main repo
commits
  e46a710/20dc9a4/212e56c.
- Round 2.1 (complete): first fleet-wide decoupling audit —
removed
  hardcoded "Helix VPN" project-name references from 19
submodules'
  governance files + helix_core's real Rust source metadata.
- Round 3 (complete): real Phase-0 engineering — helix_core WG CLI
  binaries + quinn/h3 QUIC + MASQUE framing + map.json reconciler
  (992e1be); helix_edge's first real crate, a genuine WG handshake
  traversing MASQUE-client -> edge-relay -> wg_session responder
  (08d6e18); helix_go's first real Go module, a MASQUE/CONNECT-UDP
  edge spike (57d4972, plus a .gitignore bugfix that was silently
  excluding pkg/ from version control). A second, deeper
decoupling
  audit found llm_orchestrator/vision_engine/llms_verifier/
panoptic's
  ENTIRE governance docs described an unrelated project
("HelixCode")
  — all four fixed and independently re-reviewed to a clean GO
```

(bf0ce58+ef73c3a, 2f22942, 9281cae2, 31aaceb). containers'
  compose_detector.go hardcoded an operator home directory in
  production code — fixed with a real os.UserHomeDir() resolution
+
  a regression test (a432efa). Every fix passed independent
adversarial
  code review before being accepted, iterating to a clean GO where
the
  first pass was incomplete (§11.4.125/§11.4.134).

What's next
----
1. git fetch --all --prune && git submodule foreach --recursive
'git fetch --all --prune --quiet'
2. Read docs/CONTINUATION.md fully, starting at "ROUND 3: FULLY
LANDED"
3. Nothing from Rounds 1-3 is outstanding. The one carried-forward
   follow-up round (not yet dispatched): broken
   Constitution.md self-links (should be
uppercase
   CONSTITUTION.md) in docs_chain; stale package tables in
   challenges/security; PascalCase GitHub.sh/GitLab.sh upstream
   scripts in 11 "borrowed" submodules (§11.4.29 lowercase-
snake_case
   violation).
4. Otherwise, await new operator instructions (the operator
indicated
   mvp4 work would follow full Round 2/3 completion).